

Question Paper

Exam Date & Time: 24-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU

THIRD SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Analog Electronic Circuits [ECM_2121]

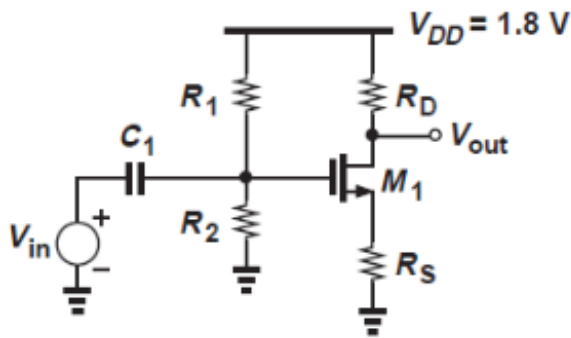
Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Design the circuit shown in figure for a voltage gain of 5 and a power budget of 6 mW. Assume the voltage drop across R_S is equal to the overdrive voltage of the transistor and $R_D = 200\Omega$. Assume $\mu_n C_{ox} = 200 \mu A/V^2$, $\lambda=0$, and $V_{TH}=0.4V$. (4)
- A)



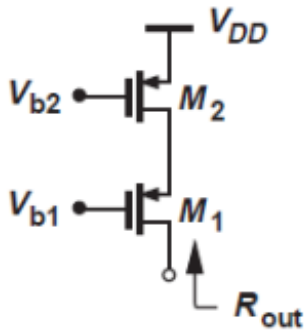
Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) An NMOS device carries 1 mA with $V_{GS}-V_{TH} = 0.6 V$ and 1.6 mA with $V_{GS} - V_{TH} = 0.8 V$. If the device operates in the triode region, calculate V_{DS} and W/L . Assume $\mu_n C_{ox} = 200 \mu A/V^2$, $\lambda=0$, and $V_{TH}=0.4V$. (3)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- C) (3)

The PMOS cascode of the circuit must provide a bias current of 0.5mA with an output impedance of 40 k Ω . If $\mu_p C_{ox} = 50 \mu A/V^2$ and $\lambda = 0.2 V^{-1}$, determine the required value of $(W/L)_1 = (W/L)_2$.

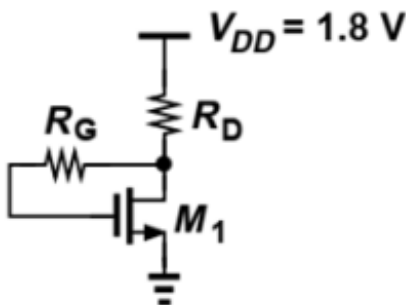


Course Outcome: "CO2" / Cognitive Level: "Application"

- 2) A MOS differential pair operated at a bias current of 0.8 mA employs transistors with $W/L=100$ and $\mu_n C_{ox}=0.2 \text{ mA/V}^2$, using $R_D=5K\Omega$ and the common-mode gain is 0.001. Find the differential gain and the CMRR. (4)
- A)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) Assume $\mu_n C_{ox} = 200 \mu A/V^2$, $\lambda = 0$, and $V_{TH} = 0.4V$ for NMOS device. The self-biased stage of the circuit must be designed for a drain current of 1 mA. If M1 is to provide a transconductance of $1/(100)$, calculate the required value of R_D . (3)



Course Outcome: "CO2" / Cognitive Level: "Application"

- C) (3)

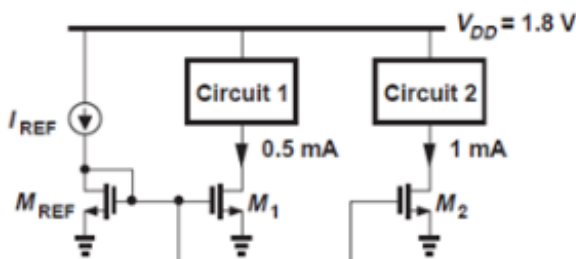
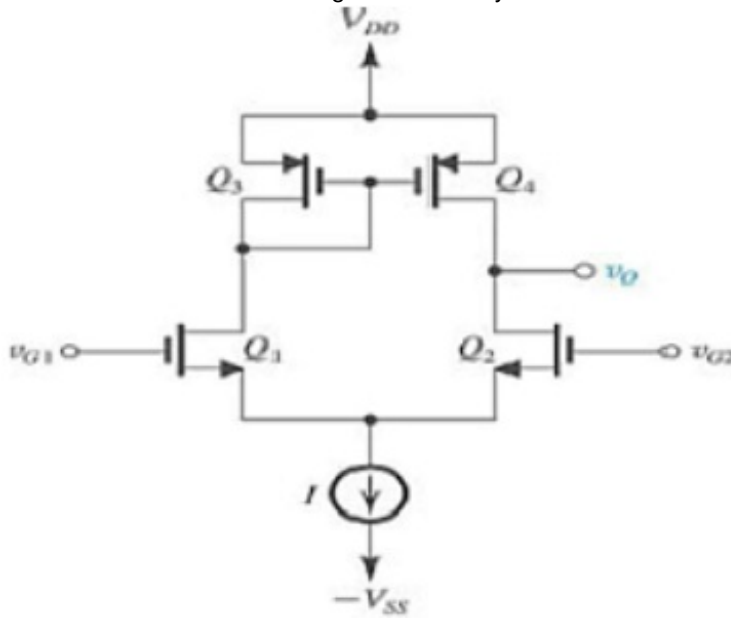


Figure shows an arrangement where M1 and M2 serve as current sources for circuit 1 and 2. Design the circuit for a power budget of 3 mW.

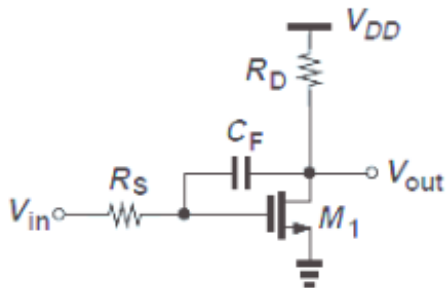
Course Outcome: "CO2" / Cognitive Level: "Application"

- 3) An active-loaded NMOS differential amplifier operates with a bias current I of $100\mu\text{A}$. The NMOS transistors are operated at $V_{ov}=0.2\text{V}$ and the PMOS dives at $|V_{ov}|=0.3\text{V}$. The Early voltages are 20V for the NMOS and 12V for the PMOS transistors. Find G_m , R_o (output R), and A_d . For what value of load resistance is the gain reduced by a factor of 2? (4)
- A)



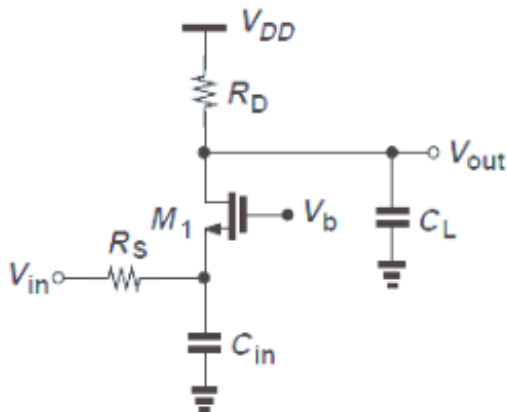
Course Outcome: "CO2" / Cognitive Level: "Application"

- B) Estimate the pole of the circuit shown. Assume $\lambda = 0$. Calculate C_{in} if $g_m = (150\ \Omega)^{-1}$, $R_D = 2\ \text{k}\Omega$, and $C_F = 80\ \text{fF}$. (3)



Course Outcome: "CO3" / Cognitive Level: "Application"

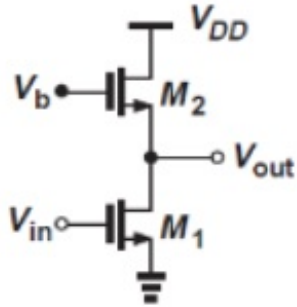
- C) Compute the poles of the circuit shown in circuit. Assume $\lambda = 0$. (3)



Course Outcome: "CO3" / Cognitive Level: "Application"

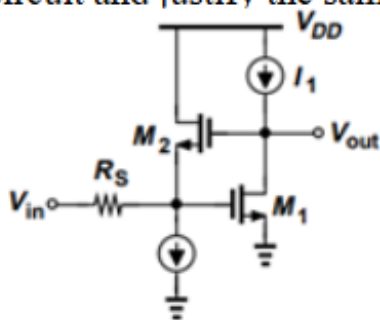
- 4) Identify the transistor capacitances and determine which ones are in parallel and which ones are grounded on both ends and compute the poles. (4)

A)



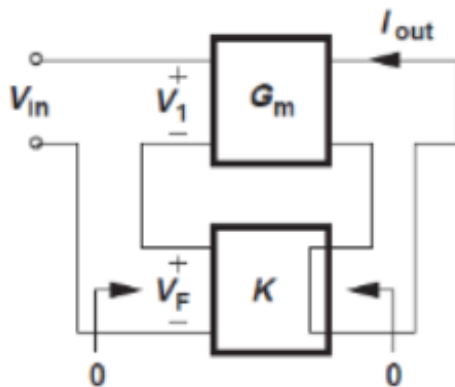
Course Outcome: "CO3" / Cognitive Level: "Application"

- B) Determine the polarity of the feedback of the circuit and justify the same. (3)



Course Outcome: "CO4" / Cognitive Level: "Understand"

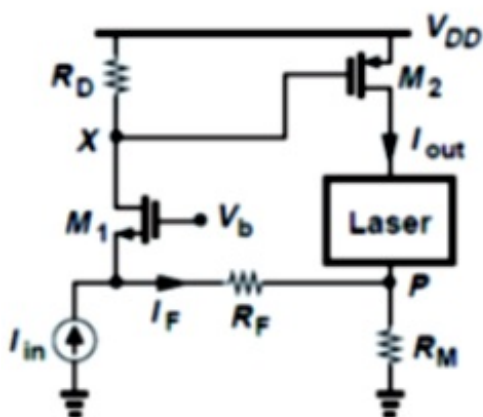
- C) Identify the feedback topology for the circuit shown and obtain the expression for closed loop gain and I/O impedances. (3)



Course Outcome: "CO4" / Cognitive Level: "Understand"

- 5) Analyze the circuit shown in figure and determine the closed-loop gain and I/O impedances if R_M is very small and R_F is very large. (4)

A)



Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Determine the expression for efficiency in Class - AB power amplifier and find its maximum efficiency. (3)

Course Outcome: "CO5" / Cognitive Level: "Understand"

- C) Explain the working principle of Hartley Oscillator with a neat circuit diagram. Also, determine the oscillation frequency, when $L_1 = L_2 = 5\text{mH}$ and $C = 10\mu\text{F}$, (3)

Course Outcome: "CO4" / Cognitive Level: "Understand"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

ECM_2121:Analog Electronic Circuits

#	Course Outcome	Description	Marks
1	1	CO 1	3
2	2	CO 2	21
3	3	CO 3	10
4	4	CO 4	13
5	5	CO 5	3
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

ECM_2121:Analog Electronic Circuits

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	5	15(30%)
3	Application	8	27(54%)
4	Analysis	2	8(16%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)